清华大学热能工程系 2007 级“燃烧学”期中考试试卷（A 卷）

Middle Term Examination of Combustion (A)

考试时间(Date)： 2010 年 4 月 26 日 班级(Class)：____

学号(Student No.)：_____ 姓名(Name)：_____

Instructions(提示):

1 请将你的学生证放在桌子的右上角（Please put your Photo student ID on the desk）

2 考试为**开卷考试**，允许带教科书，作业，课件，笔记本和另外的参考书各一样，但不能互借，答题必须独立完成。（The examination is **OPEN BOOKS**. Calculators, textbook, homework and notebook are allowed, but they can not be lent to others. You should answer the questions independently and help is not allowed）

3 试卷共有8题，你要在120分钟内完成这次考试，先检查一下你的卷子，题目是否对，共有5页。答案必须写在答题卷上，试题卷上回答无效。答题必须明确，清晰，计算题应给出单位。（There are 8 questions on this exam with 5 pages. You have up to 120 minutes to work on this exam. Please be sure that you have a complete exam package. Please work exam problems on the pages provided. Show your work neatly and write your answers. Give units for all answers requiring them.）

4 如果试卷不对，请立即与监考教师说明。（If a page is missing, take the exam to a proctor immediately.）

5 强烈推荐你先读一下所有的问题，并对回答问题的时间进行按排。答卷完成后请再仔细检查一下。（Budget your time for each question and be sure to first answer those questions that you know how to work. Check your work after completing the exam.）

6 请你现在就在答卷纸和试卷纸上写上你的**名字与学号**，并在你的答卷纸首页上标明你的试卷的类型（A或B卷）（**PRINT your name NOW** in the spaces at the top of your pages. Mark the exam type A or B）

7 答卷可以用中文或英文，如果是双语的，以中文为准。试卷有英译的，也以中文为准。(The answers can be in English or Chinese. If two languages are used in one answer, I will take Chinese as the selection. Some of the problems have been translated in English. If there are any difference between Chinese and English, Chinese is for your choice.)

试 题

1. 己烷(C_6H_{14})在干空气中燃烧，对于燃烧产物分析各组分干基组成的体积百分比为 CO2 为 8.5%、 CO 为 5.2%、O2 为 3%，其余为氮气，试确定上述燃烧过程的空燃比 A/F 和当量比 Φ？**[13 分]**

Considering hexane combustion in dry air, the dry-basis compositions of the species in product are CO2 with 8.5%, CO with 5.2%, O2 with 3%, and the rest are N2. Please determine the air-fuel ratio A/F and the equivalence ratio Φ. **[13 points]**

2. 液态的丙烷在 25℃下，用 25℃的空气进行燃烧，假设燃烧完全，反应物和燃烧产物为理想气体，**试求**下述两种情况下理想燃烧温度：**[14 分]**

(1) 当丙烷在当量比条件下燃烧；

(2) 当量比 Φ=0.4 的条件下燃烧。

The liquid Propane burns with air at 25℃. Assuming complete combustion and ideal gas, please determine adiabatic flame temperature under conditions below, **[14 points]**

(1) with equivalence ratio Φ=1.0

(2) with equivalence ratio Φ=0.4

3. 计算以下三个反应的化学反应速率，并对各反应速率的大小进行比较，假设反应温度 1800K，压力 101325Pa：**[16 分]**

$$CO + OH \rightarrow CO_2 + H \qquad (1)$$

$$H + H + M \rightarrow H_2 + M \qquad (2)$$

$$O_2 + M \rightarrow O + O + M \qquad (3)$$

其中反应(1)和(2)的正反应速率常数为

$$k_{1f} = 4.76\times10^{7}\times T^{1.23}\times\exp(-70/T) \quad \text{cm}^3/\text{mol-s}$$

$$k_{2f} = 1\times10^{18}\times T^{-1} \quad \text{cm}^6/\text{mol}^2\text{-s}$$

而反应(3)的逆反应速率常数为：

$$k_{3r} = 1.2\times10^{17}\times T^{-1} \quad \text{cm}^6/\text{mol}^2\text{-s}$$

混合物中具体摩尔分数为

$\chi_{OH}=4.0\times10^{-3}$; $\chi_{CO}=2.0\times10^{-2}$

$\chi_{H}=7.0\times10^{-4}$; $\chi_{H2}=2.0\times10^{-7}$;

$\chi_{O2}=3.0\times10^{-2}$; $\chi_{O}=1.0\times10^{-7}$;

$\chi_{N2}=0.70$; $\chi_{CO2}=0.10$;

χ_{H2O}=0.145;

Calculate the values of reaction rate of three elementary reactions below, and compare them. The temperature is 1800K, and pressure is 101325Pa：[16 分]

$CO + OH \rightarrow CO_2 + H$ (1)

$H + H + M \rightarrow H_2 + M$ (2)

$O_2 + M \rightarrow O + O + M$ (3)

The rate coefficients of forward reactions of (1), (2) are given as,

$k_{1f} = 4.76\times10^{7}\times T^{1.23}\times\exp(-70/T)$ cm^3/mol-s

$k_{2f} = 1\times10^{18}\times T^{-1}$ cm^6/mol^2-s

Meanwhile, the rate coefficient of reverse reaction (3) are given,

$k_{3r} = 1.2\times10^{17}\times T^{-1}$ cm^6/mol^2-s

The molar fraction of the mixture in product are given as below,

χ_{OH}=4.0×10^{-3}; χ_{CO}=2.0×10^{-2}

χ_{H}=7.0×10^{-4}; χ_{H2}=2.0×10^{-7};

χ_{O2}=3.0×10^{-2}; χ_{O}=1.0×10^{-7};

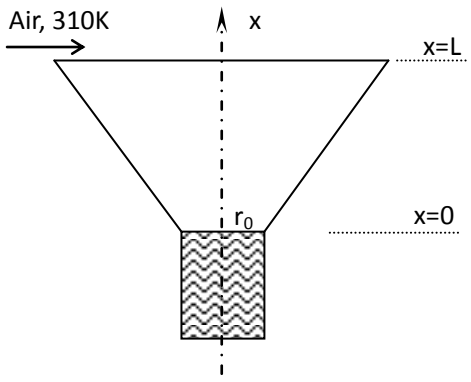
χ_{N2}=0.70; χ_{CO2}=0.10;

χ_{H2O}=0.145;

4. 如下图一个柱锥型的敞口杯中，液面始终维持不变，即 x=0 处半径 r_0=1cm，开口高度 L=5cm，倾斜角度 45°, 液体为苯(性质 ρ_l=879kg/m3; MW=78.10kg/kmol; h_{fg}=393kJ/kg；T_{boil}=353K)，空气在管口掠过，系统的环境温度 310K，质量扩散系数 $D_{C_6O_6\text{-}air}$ 为 0.93×10^{-5} m^2/s, 假设系统密度取液面处和开口端的平均，试确定：

(1) 苯的质量蒸发速率；**[9 分]**

(2) 与半径 r_0=1cm 的等直径量筒相比，二者蒸发速率的比较。**[6 分]**



In a broad-opening cylindrical/conical jar like a figure above, the interface of liquid

benzene is kept at a position of x=0, where the radius at the interface are held at r_0=1cm. The height from the interface to the top of jar are L=5cm，and the inclination is 45°. The properties of benzene include ρ_l=879kg/m3, MW=78.10kg/kmol, h_{fg}=393kJ/kg and T_{boil}=353K. The air flows across the jar with a system temperature of 310K, the mass diffusivity ($D_{C_6O_6\text{-}air}$) is 0.93×10^{-5} m^2/s. The mean density of system is obtained by averaging the value at the interface and that at jar top surface.

(1) determine the mass evoration rate (kg/s) of benzene **[9 分]**

(2) compare the evaporation rate with a 1-cm-radius (r_0=1cm) tube **[6 分]**

5．假设一个微型燃气轮机为理想的全混流反应器，进行的是甲烷-空气混和物在当量比 Φ=0.8 条件下的燃烧，假设反应器系统内温度为 1573K，如果系统压力为 5atm(即 5 个大气压)，反应器体积为 268cm^3，甲烷的流率为 0.005kg/s，试计算全混流反应器内混合物的停留时间。**[10 分]**

Assuming a micro-scale gas turbine as a well-stirred reactor, methane burns with air at Φ=0.8. The system temperature is 1573K, and the pressure are 5atm (i.e., 0.5066Mpa). The volume of the well-stirred reactor is 268cm^3. The flow rate of methane is 0.005kg/s. Please determine the residence time of mixture in this well-stirred reactor. **[10 分]**

6. 对于多组分扩散，其物理基础来自组分 *i* 的动量变化率的方程，

$$\rho Y_i \frac{Dv}{Dt} + \nabla p_i = \sum_{j=1}^{N} Z_{ij} m_{ij} (v_{j,diff} - v_{i,diff}) + \rho Y_i f_i$$

(1) 试说明方程中每项的表达物理意义，并解释所有符号的含义。**[7 分]**

(2) 表征浓度扩散的 Stefan-Maxwell 公式

$$\nabla \chi_i = \sum_{j=1}^{N} \left[\frac{\chi_i \chi_j}{D_{ij}} \left(v_{j,diff\ \chi} - v_{i\ diff\ \chi} \right) \right] \quad i = \quad 1, 2, N.$$

就是从上述组分动量变化方程中推导而来的，请问 Stefan-Maxwell 公式中 D_{ij} 物理意义上是二元扩散系数，还是多元扩散系数？**[3 分]**

As for the multi-component diffusion, its physical bass are built on the momentum changing rate of species i as below,

$$\rho Y_i \frac{Dv}{Dt} + \nabla p_i = \sum_{j=1}^{N} Z_{ij} m_{ij} (v_{j,diff} - v_{i,diff}) + \rho Y_i f_i$$

(1) please explain the physical meaning of each term or symbol in the above equation **[7 points]**

(2) the Stefan-Maxwell equation representing the diffusion due to concentration

difference (as below)

$$\nabla \chi_i = \sum_{j=1}^{N} \left[\frac{\chi_i \chi_j}{\mathrm{D}_{ij}} \left(v_{j,diff,\chi} - v_{i,diff,\chi} \right) \right] \quad i = 1, 2, ..., N$$

are just derived from the momentum equation in (1), please clarify what does D_{ij} in the Stefan-Maxwell represent among binary diffusivity or multi-component diffusivity. **[3 points]**

7. 根据燃烧理论课程所学知识，尝试搭建一个标准的柱塞流(Plug-flow)反应器系统，假设某一关键组分的质量扩散系数为 8×10^{-5} m^2/s, 同时假定反应主要发生在 2mm 的狭窄区内，尝试分析该反应器的气流流速在什么样的条件下，才能保证系统满足柱塞流反应器的假设。**[10 分]**
提示：反应器内最大的浓度梯度发生主要产生在反应区。

The knowledge during combustion study is helpful to build a standard plug-flow reactor. Assuming the mass diffusivity of the key species is 8×10^{-5} m^2/s. It is assumed that the reaction mainly occurs within a narrow zone with a length of 2mm. Please determine the proper condition for mixture velocity to ensure the basic assumption of plug flow reactor. **[10 points]**
Clue：the maximum gradient of the species concentration exists in thin reaction zone

8. 在火箭燃烧室产物中含有 H 原子和 OH 游基, 其浓度均为 $4\times10^{-6}mol/cm^3$，温度为 3000K，总气体浓度为 $4\times10^{-4}mol/cm^3$。若这些气体是在 1000K 的温度下从燃烧室排出(此时 H_2O 基本上不再离解), 此时气体密度是燃烧室内气体密度的 1/20，若排气以 3000m/s 的速度流动,试计算在下游中多远的地方能测出 H 和 OH 已有 99.9%再化合。假设除 $H+OH+M\rightarrow H_2O+M$ 外($k=10^{16}cm^6/mol^2s$, 且与温度无关), 不考虑其他任何的再化合反应。**[12 分]**
(There are H radicals and OH radicals in the combustor of rocket. The concentrations of H and OH are same as $4\times10^{-6}mol/cm^3$. The temperature of the combustor is 3000K. The total gas concentration is $4\times10^{-4}mol/cm^3$. The gas exits the combustor at the temperature of 1000K, where the formed H_2O does not dissociate. The density of the exit gas is about 1/20 of the density of the gas in combustor. Please determine the distance of the gas flow from the exit to the point of the downstream where H and OH recombined to 99.9%. The velocity of exit gas is about 3000m/s. Assume that only the recombine reaction takes place: $H+OH+M\rightarrow H_2O+M$ and $k=10^{16}cm^6/mol^2s$ and k is independent of temperature.) [12 分]